

Simulation Difficult Interview Questions

Q1:

Q1: Why do we do no timing simulations?
With timing simulations also we can validate the patterns?

Ans:- NO-timing simulations is used:
(i) to check proper netlist, lib files are read
(ii) to check if there are any internal cut points.

Timing simulations takes time and for that timing has to be first cleared by STA team.

So we use no-timing simulations to check the above cases.

Q2:

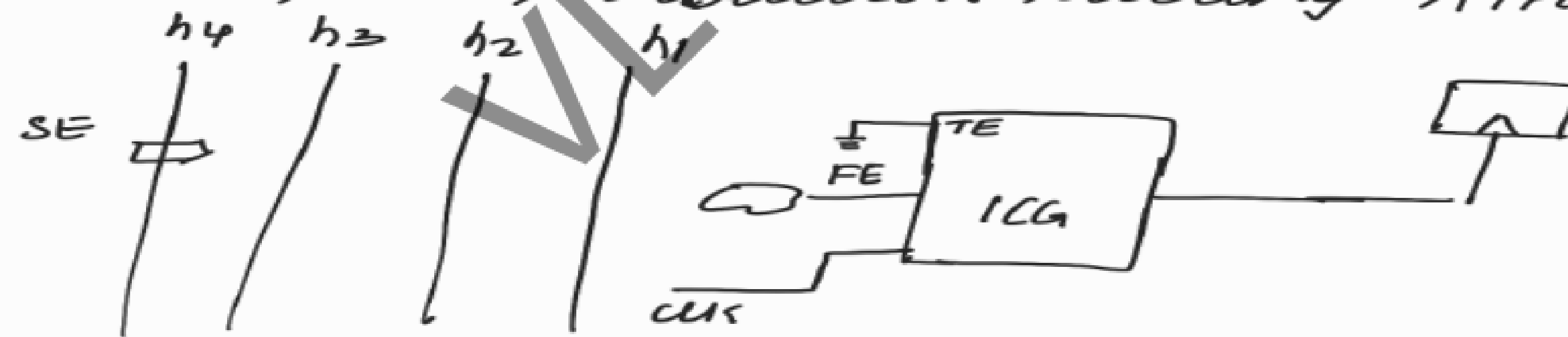
Q12 :- Why do chain pattern simulation fail?
During ATPG, we have done tracing checks.
Then why do we get simulation failure?

Ans :- Example scenario :-

Suppose we got tracing violation during ATPG because of the below reason:

If designer has manually added a new ICG to reduce dynamic power (he has done it after scan ins) and he has not informed DFT about it. He has also not controlled the TE of ICG properly.

$\therefore T3, T5 \Rightarrow$ violation during ATPG.

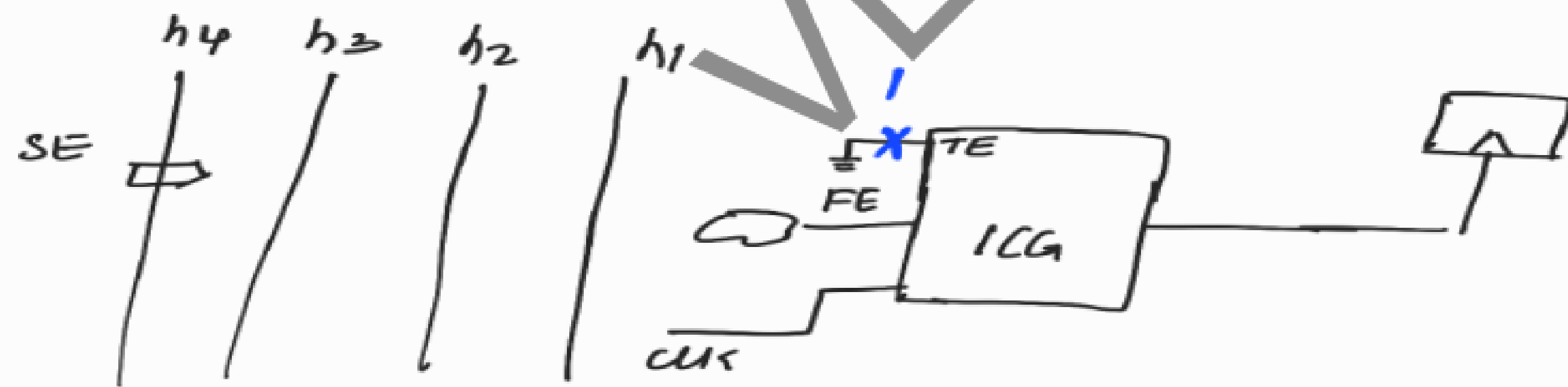


Actual fix for the tracing violation will be to connect the TE of ICG to SE.

But this ICG is present somewhere inside your design. (To connect TE of ICG to SE, you need to propagate it in multiple hierarchies)

If this has happened during temporary release, so someone has found the tracing violation by adding a cut-point & the forcing the pseudo-port to 1.

They have used this method because (,) ^{see above} is very time-consuming and fixing it using cut-point is quick.



This will pass chain tracing during ATPG.

During simulation, if we have internal cut-points, our simulations will fail.

NOTE :- For temporary releases, we can fake pass the simulation by forcing the required value.

↳ will be discussed in L3 projects

During final release (i.e. when the patterns are delivered to the tester to test the chip after manufacturing), we have to make sure that we have not used any internal cutpoints.

Otherwise the patterns will fail on the tester.

Q3:

Q13:- What is binary mismatch and x mismatch in simulation?

Ans:- Binary mismatch:- Suppose if we have taken wrong data from

simulated value
0

Expected value
1

x-mismatch:- Suppose if proper clk is not going

simulated value
X

Expected value
0

X

1

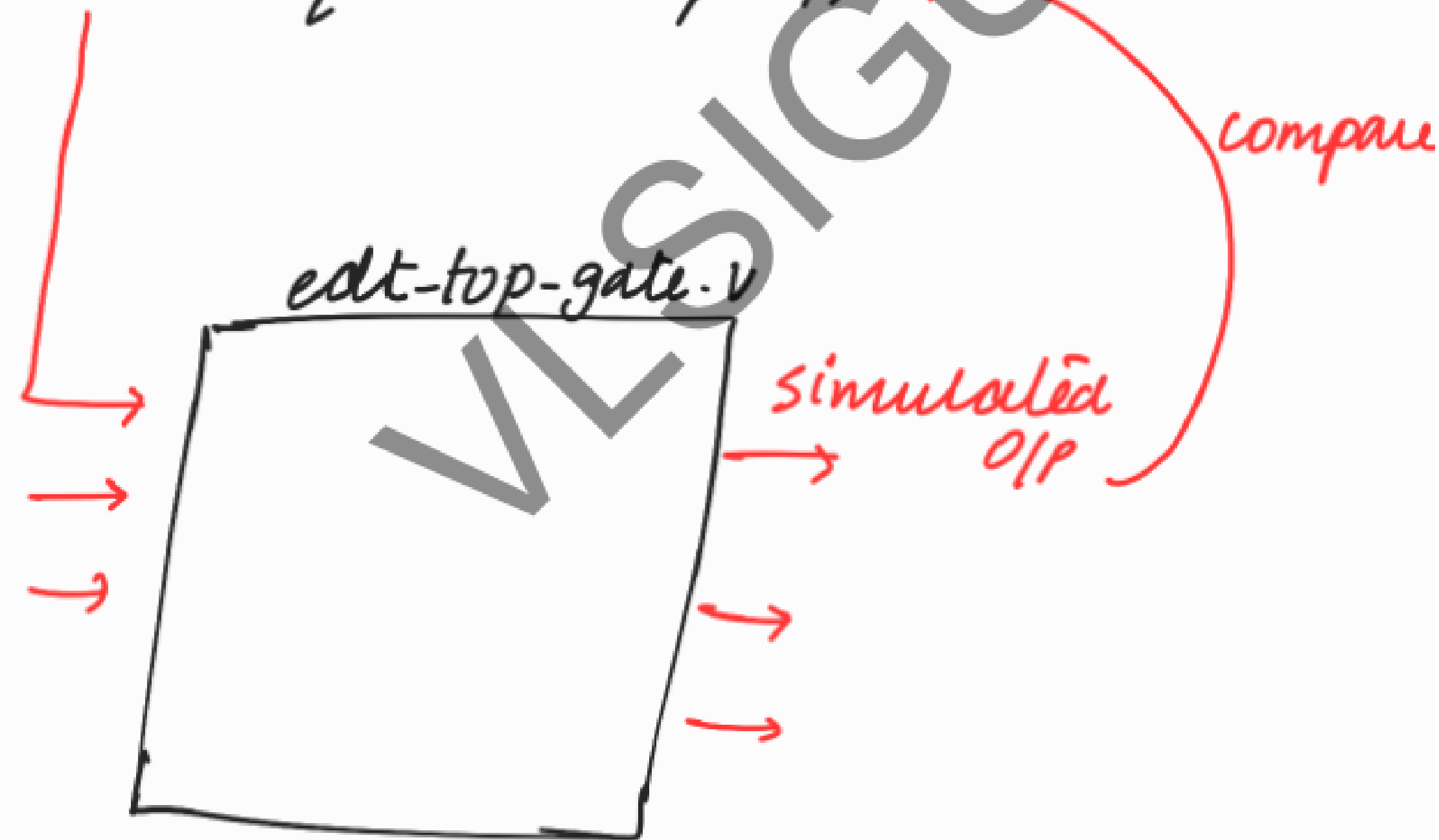
→ Reason is previous q's ans

NOTE:- Expected value:- Value written by ATPG tool
Simulated value:- Value which we get during simulated.

ATPG :- edt-top-gate.v

i/p seq + exp o/p

Sim :-



Q4:

Q20: What is unit delay simulation?

Ans:- Unit delay simulation $\rightarrow$ no timing simulation

In unit delay simulation, combinational logic delay will not be considered but in flop there will be 1 unit delay otherwise if the clock & data arrive at the same time for a flop, the flop will not sample properly.

This is no timing simulations also called as unit delay simulation.

In timing simulation, delays of combinational logic will also be considered.

Q5: Inputs and Outputs of simulation

Input: Library file, netlist, Testbench file, vector file

Output: simulation log, waveform dump

VLSIGURU